

SIRDAL-SINNES, Norway

WMO: 014310

Lat: 58.9230N

Lon: 6.9096E

Elev: 557

StdP: 94.81

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.2	-19.1	-24.5	0.4	-22.0	-21.5	0.6	-18.5	8.5	-3.3	7.8	-2.8	1.0	30	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	23.2	14.6	21.4	13.8	19.2	13.0	15.8	21.0	14.9	19.4	13.9	17.9	2.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	10.6	17.5	12.9	9.9	16.2	12.0	9.3	15.3	46.1	21.1	43.4	19.3	40.5	17.9	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	5.9	5.1	DB	-25.2	25.8	2.6	2.1	-27.1	27.2	-28.6	28.5	-30.1	29.6	-32.0	31.1	(4)
(5)			WB	-24.9	17.1	2.5	1.1	-26.8	17.9	-28.2	18.6	-29.6	19.2	-31.5	20.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-4.7	-4.7	-1.5	2.4	7.0	10.4	13.2	11.9	9.1	4.5	0.1	-3.3	(6)	
	DBStd	7.43	5.35	5.62	4.21	3.43	3.83	2.71	2.94	2.19	2.58	3.06	3.85	5.79	(7)	
	HDD10.0	2529	457	411	357	228	112	26	4	6	47	171	298	412	(8)	
	HDD18.3	5328	716	644	616	477	352	238	160	201	278	428	548	670	(9)	
	CDD10.0	243	0	0	0	1	18	38	104	63	19	1	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)	
	CDH23.3	42	0	0	0	0	2	4	34	1	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	2.3	2.6	2.5	2.6	2.2	2.4	2.4	2.1	2.1	2.2	2.3	2.3	2.3	(14)	
(15) Precipitation	PrecAvg	1967	251	182	143	102	89	99	117	157	151	229	213	234	(15)	
	PrecMax	2855	523	484	416	191	196	184	280	269	296	464	492	532	(16)	
	PrecMin	1061	21	17	6	42	9	15	33	24	32	46	28	0	(17)	
	PrecStd	394	138	126	101	38	52	49	53	53	68	113	117	158	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	6.0	10.2	15.5	23.1	23.9	26.2	22.5	19.9	14.3	9.4	6.6	(19)	
		MCWB	3.4	3.7	5.3	8.2	12.8	13.6	16.1	15.2	13.0	9.9	7.6	5.5	(20)	
	2%	DB	3.6	3.8	6.3	13.1	20.6	21.7	24.1	19.7	17.2	12.0	7.7	5.2	(21)	
		MCWB	2.5	2.2	3.4	7.3	12.2	13.0	15.3	13.9	11.6	9.0	6.5	4.2	(22)	
	5%	DB	2.6	2.8	4.6	10.5	17.6	19.3	22.3	18.1	14.9	10.5	6.3	4.0	(23)	
		MCWB	1.6	1.5	2.4	6.0	10.5	12.1	14.9	12.9	11.4	8.6	5.2	2.9	(24)	
	10%	DB	1.6	2.0	3.5	8.2	15.0	16.8	20.1	16.4	13.5	9.2	4.9	2.9	(25)	
		MCWB	0.7	0.8	1.6	4.6	9.2	10.7	13.9	12.5	10.6	7.5	3.7	1.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	4.2	5.7	9.1	14.2	15.4	17.6	16.2	14.6	12.4	8.0	5.5	(27)	
		MCDB	4.6	5.6	8.9	13.9	21.3	20.9	23.7	19.9	18.0	13.5	9.2	6.0	(28)	
	2%	WB	2.8	2.5	3.9	7.6	12.8	13.9	16.4	15.1	12.9	10.1	6.6	4.1	(29)	
		MCDB	3.5	3.5	5.8	12.2	19.1	19.5	22.0	18.4	15.5	11.0	7.7	4.9	(30)	
	5%	WB	1.7	1.7	2.9	6.4	11.2	12.7	15.5	13.9	11.8	8.9	5.3	2.8	(31)	
		MCDB	2.4	2.6	4.3	10.4	16.2	17.9	20.7	16.9	14.3	10.0	6.3	3.6	(32)	
	10%	WB	0.7	0.9	1.9	4.9	9.8	11.5	14.6	13.0	10.8	7.6	3.8	1.8	(33)	
		MCDB	1.5	1.8	3.2	7.7	14.5	16.1	19.1	15.5	13.3	9.0	4.8	2.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	8.0	7.8	8.5	9.5	9.5	9.1	8.0	7.1	6.3	5.4	6.1	(35)	
		MCDBR	5.4	6.7	8.2	13.2	15.0	15.0	14.1	11.1	10.2	8.8	6.1	5.0	(36)	
	5% WB	MCWBR	4.9	5.6	5.7	8.1	7.7	7.3	6.6	5.8	6.0	6.2	5.4	4.7	(37)	
		MCWBR	5.3	5.5	7.1	12.1	13.7	12.5	12.4	8.5	8.2	6.0	5.8	5.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.246	0.278	0.306	0.342	0.343	0.334	0.349	0.346	0.322	0.305	0.265	0.226	(40)	
		taud	2.168	2.252	2.251	2.278	2.428	2.477	2.445	2.448	2.517	2.486	2.302	2.112	(41)	
	Edn at Noon		581	750	834	861	889	902	877	854	822	725	563	467	(42)	
		Edh at Noon	45	70	95	110	102	99	100	93	74	57	40	33	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.33	0.96	2.26	3.86	4.82	5.35	4.76	3.66	2.36	1.15	0.44	0.22	(44)	
	RadStd		0.07	0.21	0.36	0.58	0.73	0.58	0.61	0.32	0.26	0.20	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-116	N/A	N/A

Nomenclature: See separate page